The mean SAT verbal score of next year's entering freshmen at the local college is 600. The college also knows 69.5% of the students scored less than 625. If the scores are normally distributed, what is the <u>standard deviation</u> of the SAT scores?

Select one:

- [ ] a. 18.25
- [x] b. 49.02
- [ ] c. 4.5
- [ ] d. 9.5

Not yet answered

Marked out of 4.00

Flag question

When finding a confidence interval for a population mean based on a sample of size 8, which assumption is made?

Select one:

- [ ] a. The sampling distribution of z is normal.
- [ ] b. The sample should be taken from a normal population.
- [ ] c. There is no special assumption made.
- [x] d. The population standard deviation ,σ is known.

Question 3

Not yet answered

Marked out of 4.00

Flag question

Question 3

Not yet answered

Marked out of 4.00

Flag question

From a sample of 200 items, 12 items are defective. The point estimate of the population proportion defective will be:

Select one:

- [ ] a. 200
- [ ] b. 0.12
- [ ] c. 12
- [x] d. 0.06

If all the points in a scatterplot lie on the equation of regression line, then the correlation coefficient (r)must be:

Select one:

- a. either 1 or -1
- b. 100
- c. -1
- d. 1

Question 5

Not yet answered

Marked out of 4.00

Flag question

A random sample of size 15 taken from normally distributed population revealed a sample mean of 75 and a sample variance of 25. The 95% confidence interval for the population mean would be closet to:

Select one:

- [x] a. (72.23 , 77.77)
- [ ] b. (74.11 , 69.88)
- [ ] c. (71.33 , 76.67)
- [ ] d. (74.33 , 75.67)

Question 6

Not yet answered

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Flag question

Question 6

Not yet answered

Marked out of 4.00

Flag question

For a certain sample, we find that

$$\sum_{i=1}^{11}(x_i-\bar{x})^2=90.$$ The sample standard deviation (s) equals

Select one:

- a. 11
- b. 9
- c. $\sqrt{3}$
- d. 3

Question 7

Not yet answered

Marked out of 4.00

Flag question

Question 7

Not yet answered

Marked out of 4.00

Flag question

A random sample of $n = 100$ yields a sample mean $(\bar{x} = 50)$. Suppose that the population standard deviation $(\sigma = 20)$. For testing

$H_0 : \mu = 45$ versus $H_a : \mu > 45$, the p-value is

Select one:

- [ ] a. 0.0062
- [ ] b. 0.05
- [x] c. 0.9938
- [ ] d. 0.5172

Question 8

Not yet answered

Marked out of 4.00

Flag question

Consider a large population with a mean of 150 and a standard deviation of 27. A random sample of size 36 is taken from this population. The standard error of the sampling distribution of sample mean is equal to:

Select one:

- a. 4.17
- b. 5.2
- c. 4.5
- d. $\sqrt{27}$

Suppose that a t-test is being conducted at the $\alpha = 0.05$ level of significance to test $H_0: \mu = 50$ versus $H_0: \mu > 50$. A sample of size 20 is randomly selected. The rejection region is:

Select one:

a. $t > 1.729$

b. $t < 1.729$

c. $t < -2.093$

d. $t > 2.093$

Question 10

Not yet answered

Marked out of 4.00

Flag question

x	0	1	2
$f(x)$	a	0.2	b

If $E(X) = 1.4$, then the value of b is

Select one:

- [ ] a. 0.4
- [x] b. 0.6
- [ ] c. 0.2
- [ ] d. 0.1

Question 11

Not yet answered

Marked out of 4.00

Flag question

Question 11

Not yet answered

Marked out of 4.00

Flag question

Suppose that $P(A) = 0.2$, $P(B) = 0.4$, and $P(\overline{A \cap B}) = 0.92$. Then the events A and B are

Select one:

- a. Dependent
- b. Independent
- c. 0.6
- d. Disjoint

Question 12

Not yet answered

Marked out of 4.00

Flag question

Question 12

Not yet answered

Marked out of 4.00

Flag question

The dean of admissions in a large university has determined that the scores of the freshman class in a mathematics test are normally distributed with a mean of 82 and a standard deviation of 8. Based on this information, what is the probability that the mean score of a sample of 64 students is at least 80?

Select one:

- a. 0.5987
- b. 0.9772
- c. 0.4772
- d. 0.0987

variables x and y

x	y
1	3
2	7
3	5
4	11
5	14

The best fitted equation line (the regression line of y on x is

Select one:

- [ ] a. $\hat{y} = 2.6 + 2x$
- [ ] b. $\hat{y} = 0.1 + 17x$
- [x] c. $\hat{y} = 0.2 + 2.6x$
- [ ] d. $\hat{y} = 0.3 + 0.5x$

Question 14

Not yet answered

Marked out of 4.00

Flag question

If the random variable X is binomially distributed with $n = 10$ and $p = 0.05$, then $P(X = 2)$ is

Select one:

- a. 0.015
- b. 0.075
- c. 0.25
- d. 0.5

Question 15

Not yet answered

Marked out of 4.00

Flag question

Question 15

Not yet answered

Marked out of 4.00

Flag question

A numerical descriptive measure calculated from the sample is called

Select one:

- a. a sampling distribution
- b. a statistic
- c. a population
- d. a parameter

Finish attempt ...